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Towards Total Recall in Industrial Anomaly Detection

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Abstract

Being able to spot defective parts is a critical component in large-scale industrial manufacturing. A particular challenge that we address in this work is the cold-start problem: fit a model using nominal (non-defective) example images only. While handcrafted solutions per class are possible, the goal is to build systems that work well simultaneously on many different tasks automatically. The best peforming approaches combine embeddings from ImageNet models with an outlier detection model. In this paper, we extend on this line of work and propose ***PatchCore****, which uses a maximally representative memory bank of nominal patch-features. PatchCore offers competitive inference times while achieving state-of-the-art performance for both detection and localization. On the challenging, widely used MVTec AD benchmark PatchCore achieves an image-level anomaly detection AUROC score of up to* 99.6%*, more than halving the error compared to the next best competitor. We further report competitive results on two additional datasets and also find competitive results in the few samples regime. Code: github.com/amazon-research/patchcore-inspection.*

1. Introduction

The ability to detect unusual patterns in images is a feature deeply ingrained in human cognition. Humans can differentiate between expected variance in the data and outliers after having only seen a small number of normal instances. In this work we address the computational version of this problem, *cold-start*[1] anomaly detection for visual inspection of industrial image data. It arises in many industrial scenarios where it is easy to acquire imagery of normal examples but costly and complicated to specify the expected defect variations in full. This task is naturally cast as a out-of-distribution detection problem where a model needs to distinguish between samples being drawn from the training data distribution and those outside its support. Industrial visual defect classification is especially hard, as errors can vary from subtle changes such as thin scratches to larger structural defects like missing components [5]. Some examples from the MVTec AD benchmark along with results from our proposed method are shown in Figure 1. Existing work on cold-start, industrial visual anomaly detection relies on learning a model of the nominal distribution via auto-encoding methods [12, 36, 44], GANs [2, 39, 43], or other unsupervised adaptation methods [42, 56]. Recently, [4, 10] proposed to leverag common deep representations from ImageNet classification without adaptation to the target distribution. Despite the missing adaptation, these models offer strong anomaly detection performance and even solid spatial localization of the defects. The key principle behind these techniques is a feature matching between the test sample and the nominal samples while exploiting the multi-scale nature of deep feature representations. Subtle, fine-grained defect segmentation is covered by high-resolution features, whereas structural deviations and full image-level anomaly detection are supposed to be covered by features at much higher abstraction levels. The inherent downside of this approach, since it is non-adaptive, is the limited matching confidence at the higher abstraction levels: high-level abstract features from ImageNet training coincide little with

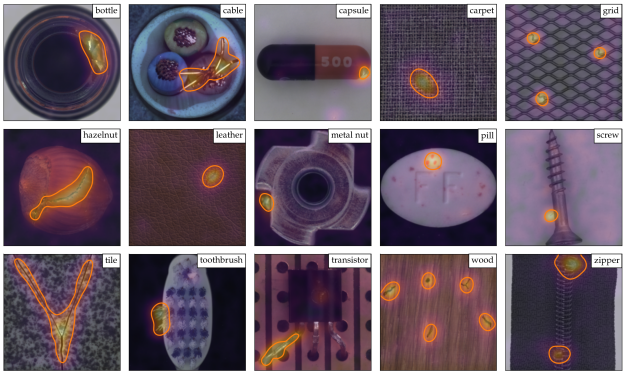


Figure 1. Examples from the MVTec benchmark datasets. Superimposed on the images are the segmentation results from *PatchCore*. The orange boundary denotes anomaly contours of actual segmentation maps for anomalies such as broken glass, scratches, burns or structural changes in blue-orange color gradients.

* Work done during a research internship at Amazon AWS.
[1]Commonly also dubbed one-class classification (OCC).